Automated Quality Metrics Assessment in Real World Data for Rare Diseases Registries EU Regulatory Compliance – AutoQ-RWD

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Abstract

Real World Data (RWD) from patient registries provide crucial evidence for rare disease care, therapy research and regulatory decision-making. However, the utility of these data depends on their quality, interoperability and compliance with regulatory standards. To address this, the European Medicines Agency (EMA) has defined a set of data quality metrics, but their adoption in the Rare Diseases (RD) field remains abstract.

In this project, we present a twofold scalable approach to operationalise EMA's quality metrics for the RD domain. First, we map the relevant elements from the EMA framework and align them with RD domain-specific needs. Second, we introduce , an automated tool that assesses these metrics using semantic technologies. The system uses the CARE-SM semantic model to structure registry data according to the FAIR principles and applies SPARQL queries to evaluate EMA quality dimensions such as accuracy, completeness, and coherence in a fully automated manner. Results are then visualised through a web-based interface, providing actionable insights while preserving data privacy.

We demonstrate the feasibility of this approach using mock data from the Euro-NMD ERN registry. By combining semantic modelling with automated quality checks, AutoQ-RWD transform data quality assessment from a manual, time-consuming process into a standardised, scalable operation. This research facilitates the evaluation of regulatory compliance by registry owners and regulatory authorities, and the provision of regulatory-grade real-world data in the qualification and registration of novel outcome measures, biomarkers and therapies for rare diseases.

Keywords

Quality Control, Rare Diseases, Semantic modelling, European Medicines Agency, CARE-SM, Euro-NMD ERN, Real-world Data, SPARQL, FAIR

1. Introduction

Real-world data (RWD) from patient registries provide essential evidence for rare disease (RD) care, therapy research, and regulatory decision-making. Particularly, they play an increasing role in post-marketing surveillance of innovative medicines for RD. However, for this data to be useful for the above-mentioned objectives, ensuring data quality is invaluable.

The European Medicine Agency (EMA) has issued provisional quality criteria for registries to demonstrate regulatory compliance[1][2][3]. These criteria define key dimensions of data quality, including accuracy, precision, traceability, completeness, coverage, conformance, uniqueness, and timeliness, and provide concrete examples of how each metric can be measured. However, the EMA guidelines stop short of providing detailed guidance on how to operationalise these criteria

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While the EMA guidelines outline important data quality metrics, they do not provide detailed methods for assessing and implementing these standards directly in existing registry data. The challenge with directly applying these criteria to a registry is that the data in most rare disease registries are often complex, unstructured and heterogeneous. This variability makes it difficult to assess quality in a consistent, automated way across different datasets. Heterogeneous data models and inconsistent use of clinical terminologies impede cross-registry analyses and often prevent alignment with other national and international initiatives. Additionally, many registries still rely on manual or non-standardised quality checks, creating bottlenecks in demonstrating compliance with EMA requirements and other registries.

This gap can be addressed by adopting the FAIR principles[4], a set of 15 principles that promote appropriate data governance. The FAIR principles are grouped into four main categories: Findability, Accessibility, Interoperability, and Reusability. Given the characteristic low prevalence of rare disease cases and limited patient data, the rare disease community has invested significant effort in implementing the FAIR principles to enhance interoperability and improve research outcomes. One of the most used FAIR approaches for representing structured rare disease data is by using semantic models.

However, while the FAIR principles improve data governance and interoperability, they do not directly address data quality. This creates an additional challenge: ensuring that FAIR-compliant registries also meet quality standards defined by frameworks such as those from the EMA, and finding ways to embed these quality assessments within FAIR workflows.

Semantic modelling allows for structurally encoding elements in a machine-interpretable format, capturing both the meaning and the values of the data using Resource Description Format (RDF) compatible formats and standard ontologies. This approach supports interoperability, scalability, automated reasoning and facilitates the application of AI algorithms and machine learning. Among the semantic models developed for rare disease registries, The Clinical And Registry Entries (CARE-SM) model stands out[5][6]. It formalises the minimal dataset required for rare disease registries in Europe, known as the Common Data Elements (CDEs), as recommended by the EU RD Platform[7]. The CARE-SM model is being adopted to integrate heterogeneous datasets into a unified framework supported by flagship European projects such as ERDERA and EJPRD[8][9].

In this article, we aim to bridge the gap between the EMA quality metrics and their practical application for rare diseases. Our approach involves two main tasks. First, we conduct a mapping effort that translates the EMA quality metrics into the context of rare diseases. This involves aligning the relevant data elements from the CDEs with the CARE-SM semantic model and domain-specific needs, making the metrics both understandable and actionable for non-technical stakeholders. This mapping provides a foundation for assessing data quality in RD, helping non-technical roles to navigate and align with other data models. By transforming registry data into a standardised, semantic model like CARE-SM, we can ensure interoperability, scalability, and consistency in the quality assessment process. This model not only helps in aligning the data with the EMA criteria but also facilitates easier comparison across different registries, even if their underlying structures vary. Therefore, adding semantics to the data is a crucial step in operationalising the EMA quality metrics and making them actionable for rare disease research.

Second, we present AutoQ-RWD, a technical implementation that operationalises these quality metrics using semantic technologies. AutoQ-RWD leverages the CARE-SM model and SPARQL queries to assess data quality in a standardised and automated manner. To demonstrate the applicability of our approach, we apply AutoQ-RWD to synthetic data from the Euro-NMD registry, illustrating its potential for integration into real-world data workflows and federated infrastructures[10].

2. Materials and methods

2.1. EMA Quality Metrics

The European Medicine Agency (EMA), through its data quality metrics report, organises quality assessment into four core dimensions[3]. Each dimension is further divided into subdimensions, which are assessed using one or more metrics groups, creating individual metrics. This multilayer approach allows for evaluation of diverse quality aspects.

The four dimensions are:

- **Reliability**: Measure how accurately data reflects real-world data values. Its subdimensions are **Accuracy**, **Precision**, and **Traceability**.
- **Extensiveness**: Ensures sufficient data points, variables, and temporal/spatial representation. Its subdomains are **Completeness** and **Coverage**.
- **Coherence**: Assessment of logical consistencies and internal agreement of data elements within the dataset. Its subdomains are **Format coherence (conformance)**, **Relational coherence (conformance)**, **Semantic coherence (conformance)**, and **Uniqueness**.
- **Timeliness**: Checks the availability of data at the right time for regulatory decision making, that in turns entails that data is collected and made available within an acceptable time.

The metrics groups for the four dimensions are:

- **Independent data checks:** Verifications without requiring additional dataset knowledge, covering issues such as null values, syntax errors, and redundancies.
- **Plausibility checks:** Detection of unrealistic or impossible data based on dataset knowledge. Examples can be medical values out of range, illogical timelines, or biologically inconsistent records.
- **Conformance checks:** Metrics evaluating whether data adheres to predefined standards for structure, dictionaries or formats.
- **Comparison to other data sources:** Metrics comparing data against trusted references sources to assess completeness and reliability.
- **Checks on dataset metadata:** Validate data against metadata, such as expected values or verifying standards.

2.2. Semantic Data Model

To semantically represent the rare disease dataset, the CARE-SM model was employed[5][6]. CARE-SM is based on the CDEs defined by the European Platform on Rare Disease Registration (EU RD Platform) and provides a structured machine-readable framework for modelling clinical data[7]. The model is formatted in RDF (Resource Description Framework) and serialised with nQuads syntax. The semantic model has been implemented using the latest stable release of CARE-SM (v1.0.0), and the pipeline used to convert tabular data into RDF is publicly available[11].

2.3. EMA Quality Metrics to Rare Diseases and CARE-SM mapping

For the mapping between EMA metrics to Rare Diseases field with CARE-SM, each metric has been mapped to specific CARE-SM properties based on the data elements required for evaluation. The matrix used for the mapping is given in Table 1.

Where semantic gaps existed, logical approximations or derived mappings were used. This semantic mapping ensures that the quality metrics are machine-readable, reproducible, and adaptable to other rare disease registries using the CARE-SM schema.

Metric selection was guided by three criteria: (1) Plausibility in rare disease contexts, (2) data structural integrity (Ex: ontology compliance, syntax, etc.), and (3) technical feasibility (Ex:

availability of metadata, queries to other sources, etc.). Metrics failing any criteria were excluded. Metrics failing any of the above criteria were excluded from implementation.

The mapping process of the metrics involved contextual adjustments. This can be in the form of terminology mapping or specifying the metric. For instance, when mapping the conformance check from the Coherence dimension, the question has changed from looking relevant variables that employ code lists according to external standards to percentage of phenotypes coded with HPO[12]. To enhance the specification of metrics for rare diseases, other contextual adjustments were made, including checking specifically for missing CDEs values when addressing the missing values metric.

2.4. Use case Dataset

The dataset used to populate the semantic model was derived from European Reference Network for Neuromuscular Diseases (Euro-NMD ERN) registry, which collects clinical data from patients with neuromuscular disorders across expert centres within the ERN Euro-NMD network[10]. A synthetic dataset was created based on the Euro-NMD Registry Codebook, with 300 fictional patients and 325 variables. The data is available in Zenodo[13].

To ensure sufficient population of the semantic model, synthetic data was generated using a Python script that randomly selects values from predefined lists defined in the code[14]. While the data maintains structural validity, it does not carry clinical significance and is intended solely for testing the system's functionality and performance.

This test data has been used in the results section to show better the different EMA metrics adapted in this article.

2.5. Graph Database Deployment

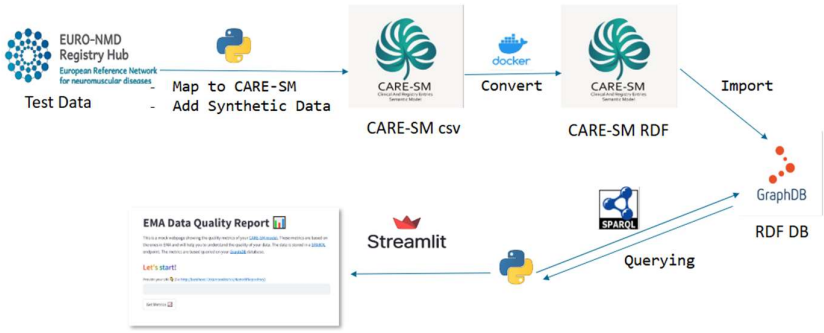
The RDF dataset was deployed in GraphDB v10.8.4, a semantic graph used for storing and querying RDF data. The repository was configured with enabled inference, to support relations between ontologies, and context indexing. All SPARQL queries used for assessing data quality metrics were executed to be compatible with any CARE-SM dataset and can be found in the Supplementary Table 2[15].

2.6. Web-based Quality Assessment Tool

To facilitate user interaction and visualisation of the data quality metrics, a web-based interface was developed using Streamlit (v1.45.0) library in Python (v3.12.3). Visualisations include bar plots generated with Streamlit's native plotting tools and interactive pie charts created using Plotly (v6.1.2). Users start by entering the URL of their semantic database endpoint into the application. The backend then sends POST requests containing predefined SPARQL queries to the endpoint, retrieves the results, and renders them in real time. The code is stored in Github[16].

A schema of the whole workflow interaction can be seen in Figure 1.

Figure 1. Workflow from test data to the quality report in a webpage.



3. Results

3.1. EMA Quality Metrics to CARE-SM model

The mapping of EMA metrics to rare disease fields within the CARE-SM framework was largely successful, with each relevant metric mapped to specific CARE-SM properties based on the data elements required for evaluation. The mapping matrix, which outlines these connections, is presented in Table 1. However, approximately half of EMA metrics were ultimately excluded from the final mapping. This was due to various reasons, including the fact that several metrics could not be generalised across rare disease datasets, making their implementation inconsistent in the rare disease context. Other metrics were theoretically adaptable to the rare disease field but were ultimately not feasible for technical mapping due to limitations related to data availability or structure. In multiple cases, required metadata was missing and no alternative data sources were available to compensate for these gaps. Additionally, some definitions could be aligned with genetic variables; however, the necessary data structural integrity was not yet in place to support their inclusion. As a result, these metrics did not meet the established mapping and feasibility criteria and were excluded from the final dataset. For instance, certain metrics did not have available metadata or could not be queried from the available data source, preventing their inclusion in the final mapping. These metrics were excluded to ensure that only those with full technical and data integrity could be implemented within the CARE-SM model. The excluded metrics and its reasons can be found in Supplementary Table 1[15].

Table 1. Mapping Metrics from EMA to Rare Diseases CARE-SM. *The first three columns are the EMA quality subdomains, metrics and their definitions. The fourth column is the mapping of the EMA definition to the rare disease field. And the last column is the CARE-SM property where the rare disease metric will be search.*

EMA Subdimension	EMA Metric	EMA definition	Rare Disease Definition	Rare Disease – CARE-SM property
Accuracy	Plausibility checks	Number and percent of records where data values don't agree with standards or knowledge or feasible ranges	% Patients with Disability Score above a certain threshold	Disability
Accuracy	Plausibility checks	Number and percent of records where values of repeated measurement of the same fact don't show expected variability	% Changes of Disability in a set of days	Disability
Accuracy	Plausibility checks	Number and percent of records where	% Duplicate Patient IDs	All variables

		duplicate records aren't flagged		
Precision	Independent data checks	The number of decimal points used in data values, and their distribution	The number of decimal points used in data values, and their distribution	Corporal
Completeness	Independent data checks	Percentage of records for which data are missing for a given variable	CDE properties with missing values	All variables
Completeness	Independent data checks	Percentage of patients who have a certain number of measurements for a given variable	% Patients with multiple disability checks (No Disability, Multiple Checks and One check)	Disability
Format Coherence (conformance)	Conformance checks	For relevant variables, % of records where data values conform to allowable values or ranges	% HPO terms in the Phenotype property % NCIT terms in the Sex property	Phenotype Sex
Relational Coherence (conformance)	Conformance checks	For computed values, % of records where computed values conform to programming specifications	Check if the age is correct based on the date of birth and the date of the measurement.	Birth Date
Semantic coherence (conformance)	Conformance checks	For relevant variables which employ code lists according to external standards, % of patient records which use a given code list	% ORDO terms in the Diagnosis property	Diagnosis;
Currency (Is your data acceptably up to date?)	Independent data checks	Average time of updates in a database	Avg. time between even Id (each time you go to a visit)	All models

Of the 21 EMA metrics analysed, 10 (47.61%) have been successfully adapted to the context of rare disease registries with CARE-SM. These metrics span all four EMA dimensions: Reliability, Extensiveness, Coherence, and Timeliness. Metrics were selected based on their relevance to rare disease data structures and technical feasibility within current registry implementations.

The highest coverage has been achieved by the Timeliness dimension, with 100% mapping of the single metric assessing whether the data is up to date. The Extensiveness and Coherence dimensions each had half of their metrics mapped, with 2 out of 4 and 3 out of 6, respectively. Extensiveness metrics mapped are in the Independent Data check metrics group, as they can be queried technically. It is not possible to map the metrics comparing our data to other data sources, as this mapping exercise did not involve multiple datasets. Metrics from the Conformance Check metric group, which assess the correctness of the data within the database, were mapped to the Coherence dimension. The other metrics haven't been mapped as they do not fit semantically to any Common Data Element.

The metric with the least coverage metric is Reliability, with 40% (4 out of 10 metrics). The metrics related to the Accuracy sub-dimension are focused on the values of the data, assessing their standards and variability. The metrics that are not mapped are related to metadata, comparing it with other sources or having specific knowledge of the dataset. A summary of the percentage of mapped dimensions can be found in Table 2.

Table 2. Percentage of mapped dimensions.

Dimension	Mapped	Total	Percentage
Reliability	4	10	0.4
Extensiveness	2	4	0.5
Coherence	3	6	0.5
Timeliness	1	1	1

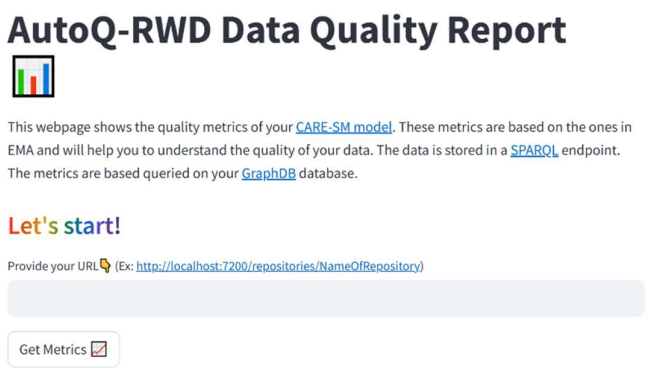
The next section describes the results of the web interface tool and the queries done within it, organised per EMA metric sub-dimensions.

3.2. AutoQ-RWD Assessment Tool

The AutoQ-RWD system demonstrates the implementation of EMA-derived data quality metrics within a semantic infrastructure for rare disease datasets[13]. The tool was deployed locally and tested using a mock dataset derived from the Euro-NMD registry. Researchers can integrate the tool into their own infrastructure by hosting the Streamlit-based interface and linking it to their data repository or FAIR Data Point[17].

A typical user journey begins with launching the web interface and entering the URL of their RDF database into the initial input field, as illustrated in Figure 2. Upon submission, the backend initiates a series of SPARQL queries to the specified endpoint, retrieving and visualising the results in real time. The metrics presented in this Results section are based on the synthetic Euro-NMD.

Figure 2. 'Welcome' webpage. *In order to run your database, you need to add your link to the grey textbox and click 'Get Metrics'.*



The system supports three types of result representations: Dynamic Threshold, Static Result and Interactive Plot. Dynamic Threshold metrics allow users to adjust input parameters, such as thresholds, which directly influence the metric outcome. Static Results are computed once at the beginning of the assessment and remain unchanged. Finally, Interactive Plots provide visual representations that improve interpretability, with features such as hover-over tooltips displaying underlying values and percentages.

3.2.1. Accuracy Metrics

Accuracy metrics, a subdomain of Reliability in the EMA metrics, assess the accuracy of data values and their variability. Two of the three metrics in this category are presented as Dynamic Thresholds, allowing users to refine the assessment criteria.

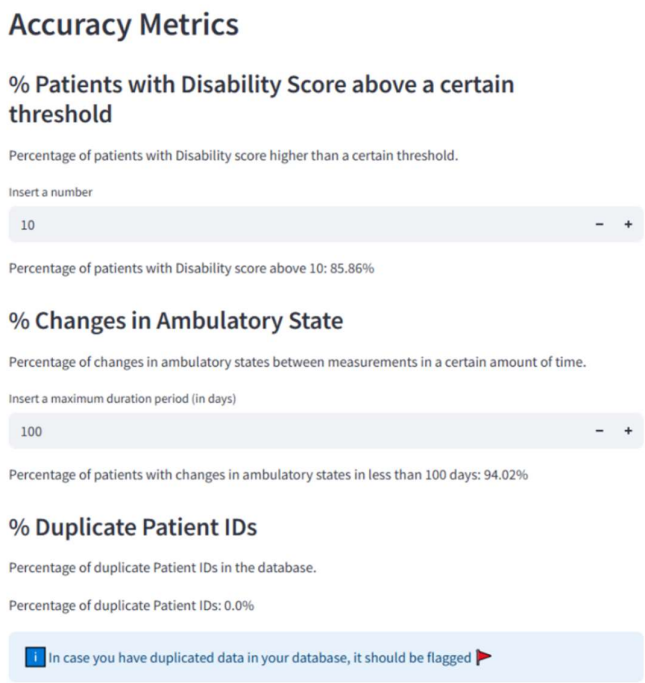
In the test data, 85.86% of patients have a Disability Score higher than 10. This metric compares the distribution of disability scores between the source data and the semantic model. Additionally,

this metric can flag score exceeding the maximum allowable value of 100, indicating data entry errors.

The metric '% Changes in Ambulatory State' evaluates the frequency of new event entries over a certain number of days. In the synthetic dataset, 7.41% of patients have a new event ID recorded every two days, suggesting high variability in clinical measurements. This metric helps identify inconsistencies or rapid changes that may affect data reliability.

The '% Duplicate patients ID' metric identifies repeated identifiers within the dataset. In the use case, no duplicate IDs were found. While duplicates do not necessarily indicate incorrect data, they must be flagged as duplicated. This information is displayed in a dedicated annotation box below the metric. The whole section can be seen in Figure 3.

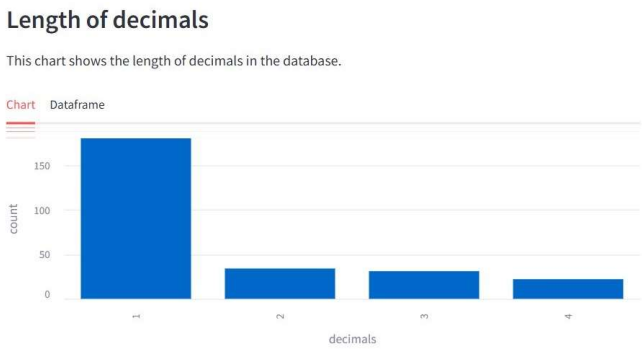
Figure 3. Display of the Accuracy Metrics section in the webpage. *To change the threshold number, you can type it or use the '+' and '=' sign.*



3.2.2. Precision Metrics

Precision is assessed through an interactive bar plot (Figure 4) showing the distribution of decimals across the dataset. Users can also view the underlying table with data values in a separate tab. In the Euro-NMD dataset, 67.4% of numerical values are recorded in the tenth form (0.1) and other entries can be found extended to hundredths (0.01), thousandths (0.001), and ten-thousandths (0.0001). This variability is mainly attributed to medication dosage values, which require high precision.

Figure 4. Barplot of Length of decimals

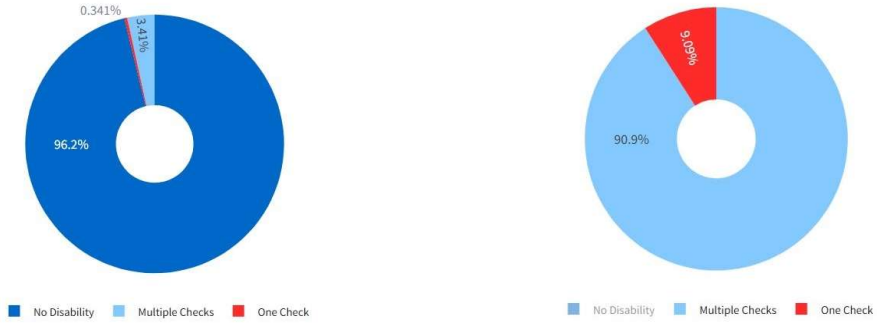


3.2.3. Completeness Metrics

In the completeness metric the system visualises missing values per Common Data Element (CDE), with red warning boxes indicating unmapped or absent properties. In the use case, all expected properties are present except for the Consent field, which was not found in the RDF model. This alert mechanism helps users identify which properties haven't been mapped to the model.

Another metric, '% Patients with multiple disability checks' is displayed as an interactive pie chart. Users can toggle visibility of chart segments via the legend. In the dataset, 96.2% of patients lack a recorded disability score, while 0.34% have undergone one assessment and 3.4% have had multiple assessments (Figure 5a). When excluding patients without any score, 90.9% of those with disability data have had multiple checks, and 9.1% have had only one (Figure 5b).

Figure 5. *In the left (a), there is the overall pie plot for all the patients with multiple disability checks. If you click a variable in the legend, you can hide one variable and get the image from the right (b).*



3.2.4. Coherence Metrics

Coherence is evaluated across three subdimensions: Format, Relational and Semantic.

Format coherence checks whether properties are annotated with mandatory ontologies. In CARE-SM, HPO terms are required for the Phenotype property and NCIT terms for the Sex property. In the use case, both properties achieved 100% compliance.

Relational coherence assesses the consistency of age values by comparing the calculated age (based on birth date and event date) with age reported in the source dataset. In the test data, 57.7% of age values were found to be correct (difference between age at visit minus the birth date compared to the variable Age according to CARE-SM), indicating potential issues in date formatting or data entry.

Semantic coherence verifies the use of recommended ontology terms. For the Diagnosis property, ORDO terms are expected[17]. In the use case, 91.58% of diagnoses were annotated with valid ORDO terms. The remaining cases correspond to undiagnosed patients, which are not represented by the Orphanet ontology in the dataset. It is recommended to model such cases using the ORPHA:616874 code ("Rare disorder without a determined diagnosis after full investigation").

3.2.5. Currency metrics

Currency metrics assess the temporal consistency of the dataset. The system calculates the average time between event entries, here defined as clinical visits. In the Euro-NMD test dataset, the average interval is 32.67 days, suggesting a monthly follow-up pattern.

4. Discussion

This study presents EMA-derived RD quality metrics and a scalable technical implementation of these metrics using semantic web technologies and RD community standards. Implementing the CARE-SM model—established as the de facto ERDERA standard and encompassing the EU CDEs—enables the developed software, AutoQ-RWD, to operate with minimal adaptation, removes the need for each registry to independently adapt and implement EMA quality criteria, and provides substantial efficiency gains through multi-registry dashboarding and benchmarking capabilities.

AutoQ-RWD, enables data owners and controllers to assess and improve the quality of their datasets in a real-time structured and reproducible manner. Beyond internal quality control, the tool also supports transparency and shareability by allowing users to expose their webpages (as servers) publicly, making them complementary pages along with a FAIR Data Point[17].

The results obtained from the synthetic Euro-NMD dataset shows the tool's ability to compute and visualise metrics across five key dimensions: accuracy, precision, completeness, coherence, and currency. For example, the system identified a 57.7% match in relational coherence for age values, highlighting potential inconsistencies in its property. In addition, we foresee improvements in metrics calculations such as precision where the length of decimal should be for each data element, instead of for the whole dataset.

The implementation in Python using Streamlit ensures accessibility and ease of customisation. Institutions can adapt their interface to their branding or add new metrics by modifying the Markdown-based layout. However, since Streamlit is primarily designed for rapid prototyping, its scalability and security limitations must be carefully evaluated before public or production deployment. These considerations will need to be addressed in future implementations, for instance by placing the interface behind a secure reverse proxy or separating the user interface from the metric-computation backend. Moreover, the SPARQL queries remain the most technically demanding component, requiring domain-specific expertise in semantic querying. This may limit adoption among non-specialist users unless supported by training.

Although the system demonstrates strong capabilities, several considerations remain. A key prerequisite for using the tool is mapping a registry's codebook to the CARE-SM model. While this step requires manual effort, the mapping developed for Euro-NMD is not only a starting point but also a reusable resource. It provides a template for registries already aligned with Common Data Elements (CDEs), significantly reducing onboarding time and accelerating adoption of EMA quality metrics. This reuse potential extends to ERDERA resources, fostering consistency across European registries.

Another challenge—common to the entire life sciences domain rather than this study alone—is the limited adoption of semantic technologies, often due to resource constraints and technical expertise gaps. Currently, the tool supports data modelled in CARE-SM, which narrows applicability. However, ongoing work within ERDERA will expand compatibility to widely used standards such as OMOP CDM and HL7 FHIR [19][20]. Integration into the ERDERA Virtual Platform will provide a centralised, FAIR-compliant environment where registries can seamlessly adopt quality assessment

workflows alongside other interoperability tools[21]. This not only simplifies onboarding but also promotes harmonisation across European rare disease registries, ensuring that data quality monitoring becomes an integral part of the broader ecosystem for research and policy support.

Finally, the tool's contribution to the FAIR principle's implementation, especially in terms of interoperability and reusability, positions it as a valuable resource for registry developers, data stewards and regulatory bodies. By enabling standardised, automated quality assessments, AutoQ-RWD supports harmonisation efforts across European rare disease registries and improves data readiness for research and policymaking.

5. Conclusion

This work demonstrates how EMA-derived data quality metrics can be operationalised for rare disease registries through semantic web technologies and community standards. By leveraging the CARE-SM model and RDF-based representations, we enable automated, standardised quality assessments that were previously manual and resource-intensive. The approach ensures interoperability, supports dynamic interaction, and provides actionable insights for both technical and non-technical users.

Our proof-of-concept with synthetic Euro-NMD data highlights the feasibility of computing and visualising metrics across five key EMA's dimensions within an intuitive web interface. This aspect positions the tool as a reusable and scalable solution for improving data reliability in rare disease research.

Looking ahead, future developments will focus on integrating real-world registry data, extending compatibility to widely adopted standards such as OMOP CDM and HL7 FHIR, and deploying the tool within the ERDERA Virtual Platform and FAIR Data Point. These steps will strengthen harmonissation efforts across European registries and embed data quality monitoring as a core component of FAIR-compliant infrastructures.

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Declaration on Generative AI

During the preparation of this work, the author(s) used Microsoft 365 Copilot in order to: Grammar and spelling check. After using these tool(s)/service(s), the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication's content.

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